Project Proposal

Secure Multi-Tenant Hosting Platform on Hetzner

Project Type	Infrastructure / Cloud Hosting Platform
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Status	Draft

1. PROJECT OVERVIEW

Description

Small organisations and individual clients often need reliable web hosting without the complexity and cost of large cloud providers. This project designs and implements a secure and scalable hosting environment on a Hetzner dedicated server. Proxmox VMs are used to organise client workloads, with multiple clients hosted per VM and Coolify managing deployment and service isolation. Cloudflare provides DNS management and security features, while an external monitoring solution based on a Raspberry Pi ensures independent availability checks via the Hetzner API.

2. PROJECT GOAL & VALUE

Goal

To deploy a secure multi-tenant hosting environment capable of running multiple client websites across Proxmox VMs, managed through Coolify with Cloudflare and external monitoring.

Problem Being Solved

Many small projects need a reliable and structured way to host websites. Setting up such an environment from scratch requires knowledge of security, virtualisation, and monitoring — which can be challenging for less experienced teams.

Value

The project demonstrates practical skills in virtualisation, containerisation, deployment management via Coolify, networking, and infrastructure monitoring.

3. PROJECT SCOPE

In Scope

– Host multiple client websites on a Hetzner dedicated server.
– Use Proxmox to provision and manage VMs, with multiple clients per VM.
– Use Coolify to manage deployment, service isolation, and container orchestration.
– Integrate Cloudflare for DNS management, DDoS protection, and security rules.
– Implement external monitoring on a Raspberry Pi using the Hetzner API.
– Provide documentation for deployment and maintenance.

Out of Scope

– Personal or local servers — all hosting and deployment uses Hetzner infrastructure.

4. FUNCTIONAL REQUIREMENTS

- Provisioning and management of VMs using Proxmox, hosting multiple clients per VM.
- Deployment and management of client services using Coolify.
- DNS configuration and security rules via Cloudflare.
- External monitoring to check server availability and basic resource usage.
- Secure SSH access and firewall configuration.

5. TECHNOLOGY STACK

Component	Technology
Virtualisation	Proxmox
Deployment & PaaS	Coolify
Containers	Docker (managed via Coolify)
Web Server	Nginx (via Coolify reverse proxy)
DNS & Security	Cloudflare
Monitoring	Raspberry Pi · Hetzner API · Custom Scripts
Infrastructure	Hetzner Dedicated Server
Dev Tools	Git · SSH

Coolify acts as the central deployment layer, simplifying multi-client service management across VMs while maintaining strong isolation and observability.

6. PROJECT PLAN

Phase	Description	Hours
Phase 1	Planning and server setup	**6 h**
Phase 2	Proxmox setup and VM provisioning	**10 h**
Phase 3	Coolify installation and service deployment	**10 h**
Phase 4	Monitoring, testing & documentation	**12 h**
Total		**38 h**

7. RISKS & CHALLENGES

- Complexity of multi-layer virtualisation
- Security misconfiguration risks
- Limited hardware resources

8. SUCCESS CRITERIA

- Multiple client websites run reliably and independently.
- Monitoring system correctly reports availability and resource status.
- Secure access and DNS configuration are verified.
- Clear documentation enables repeatable deployment and maintenance.